



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Charge and Current — $Q = It$

One equation, one trap. The trap is the units.

Key concept

CURRENT IS A FLOW OF CHARGE

Current is not “electricity”. It is the **rate at which charge flows** — how much charge goes past a point each second.

$$\text{charge flow} = \text{current} \times \text{time} \quad Q = It$$

Q = charge flow, in **coulombs (C)** · I = current, in **amps (A)** · t = time, in **seconds (s)**

Rearranged: $I = \frac{Q}{t}$ and $t = \frac{Q}{I}$

The trap, and it is the one that costs marks. t must be in **seconds**. Real questions give you *minutes*, or hours. Convert **first**, and write the converted number down so a marker can see it.

1 minute = 60 s · **2 minutes** = 120 s · **1 hour** = 3600 s

Worked example

A current of 0.25 A flows for **3 minutes**. Calculate the charge that flows.

Convert first. $t = 3 \times 60 = 180$ s $Q = It = 0.25 \times 180 = \mathbf{45\ C}$

Using $t = 3$ would give 0.75 C — sixty times too small, and no method mark for a conversion you never wrote down.

1. State what is meant by *electric current*. [1]

The rate of flow of electrical charge.

2. Write down the equation that links charge flow, current and time. [1]

charge flow = current $\times$ time ($Q = It$)

3. A doorbell draws a current of 0.85 A for 40 seconds. Calculate the charge flow. **Give your answer to 2 significant figures.** [3]

$Q = It = 0.85 \times 40 = 34\ C = \mathbf{34\ C}$ (2 s.f.)

4. A current of 1.75 A flows in a heater element for 80 seconds. Calculate the charge flow. [2]

$Q = It = 1.75 \times 80 = \mathbf{140\ C}$

5. There is a current of 0.045 A in an LED for 240 seconds. Calculate the charge flow. [2]

$Q = It = 0.045 \times 240 = \mathbf{10.8\ C}$

6. A charge of 624 C flows through a motor in 130 s. Calculate the current. [2]

$I = Q/t = 624/130 = \mathbf{4.8\ A}$

7. A phone charger supplies a current of 0.90 A for **2 minutes**. Calculate the charge that flows. [3]

$$t = 2 \times 60 = 120 \text{ s.} \quad Q = 0.90 \times 120 = \mathbf{108 \text{ C}}$$

8. A charge of 1500 C flows through a lamp in **5 minutes**. Calculate the current. [3]

$$t = 300 \text{ s.} \quad I = 1500/300 = \mathbf{5.0 \text{ A}}$$

9. A torch bulb carries a current of 0.24 A. How long must it be switched on for 72 C of charge to flow? Give your answer **in minutes**. [3]

$$t = Q/I = 72/0.24 = 300 \text{ s} = \mathbf{5 \text{ minutes.}}$$
 Convert back — the question asked for minutes.

10. A student says: “A bigger current always means more charge has flowed.” Explain why this is not necessarily true. [2]

*Charge depends on current **and** time. A small current flowing for a long time can move more charge than a large current flowing for a moment.*

11. A single closed loop carries a current of 2.0 A. Explain why the same charge flows past every point in that loop in one minute. [2]

The current is the same at every point in a single closed loop, and the time is the same, so $Q = It$ gives the same value everywhere. Charge is not used up as it goes round.